

IOWA STATE UNIVERSITY
ELECTRICAL AND COMPUTER ENGINEERING

EE 230
LAB 9 REPORT

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Introduction

In this lab, we started working with MOSFETS. At first, we used the parameter analyzer to get the V_T and K values from the slopes and intersection points on the graph it generated. Then, we built 4 different MOSFET circuit configurations, 2 in saturation and 2 in triode region, we measured i_d , V_{DS} and V_{GS} for each circuit and then compared them with our calculated values. Then we built two MOS inverter circuits and measured the output voltage for them across 25 different inputs and acquired their plots. For the 4th experiment, we built a circuit which used a MOSFET as a switch and also included a comparator. Finally, we observed how a MOSFET behaves on the oscilloscope when it is used as an amplifier.

Parameter Analyzer

To begin, the V_T and K values for an NMOS and PMOS in the CD4007 array were computed. These values are presented in the below table. We decided to use **QxN** and **QxP** for the NMOS and PMOS devices, respectively.

Device	Calculated V_T	Calculated K
CD4007 NMOS	+1.27 V	$4.51 \cdot 10^{-4}$
CD4007 PMOS	-1.47 V	$5.57 \cdot 10^{-4}$

Next, $i_D - v_{DS}$ curves were obtained for each of these devices. These are shown below. The NMOS and PMOS devices for the CD4007 were tested with six v_{GS} values ranging between 0 V and 6 V.

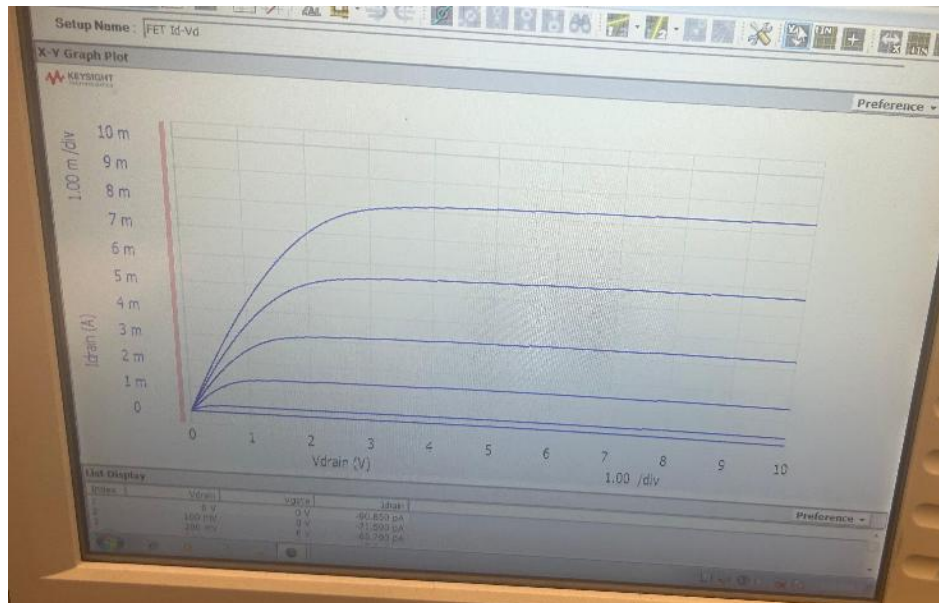


Figure 1: i_D vs. v_{DS} for an NMOS in the CD4007 array

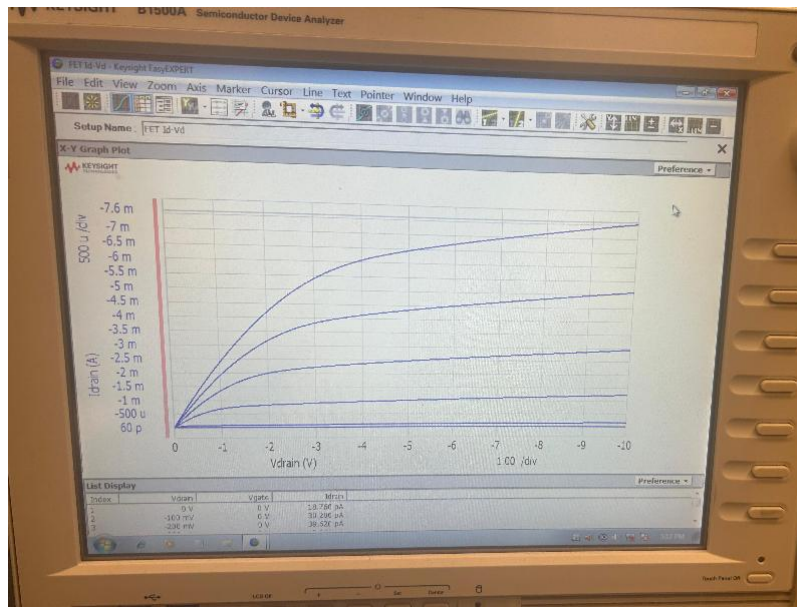


Figure 2: i_D vs. v_{DS} for a PMOS in the CD4007 array

i_D vs. v_{DS} curves are convenient for visually confirming our transistors work, but are not great for characterizing them. So, we also made $\sqrt{i_D}$ vs. v_{GS} plots for both of the transistors.

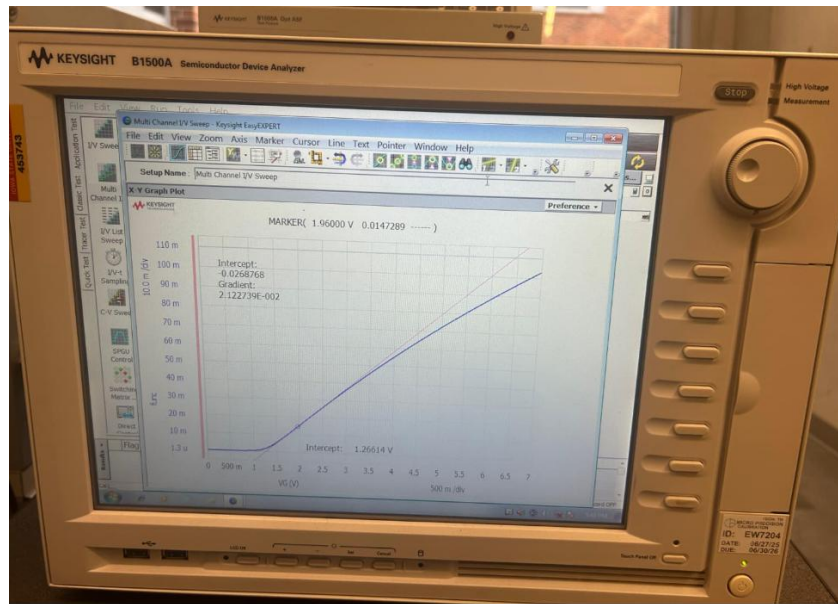


Figure 3: $\sqrt{i_D}$ vs. v_{GS} for an NMOS in the CD4007 array

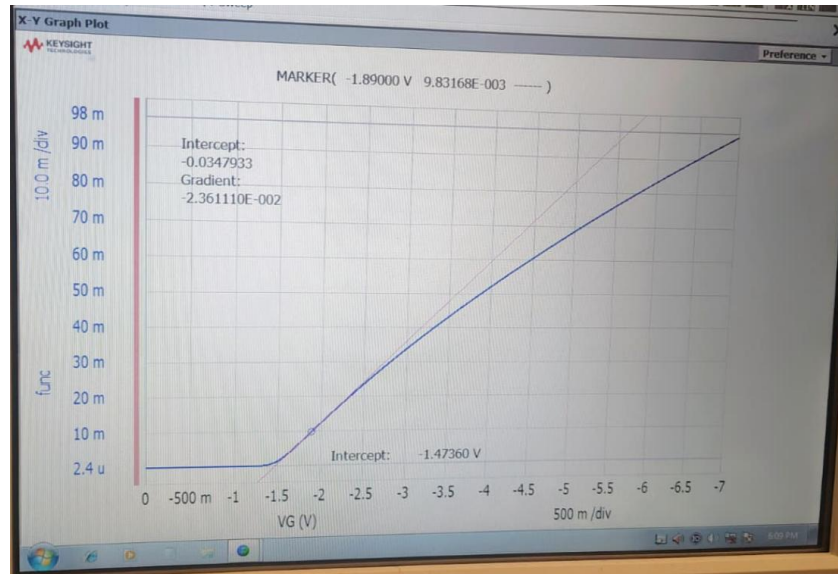


Figure 4: $\sqrt{i_D}$ vs v_{GS} for a PMOS in the CD4007 array

For a MOSFET operating in saturation:

$$i_D = K(v_{GS} - V_T)^2$$

Taking the square root:

$$\sqrt{i_D} = \sqrt{K}(v_{GS} - V_T)$$

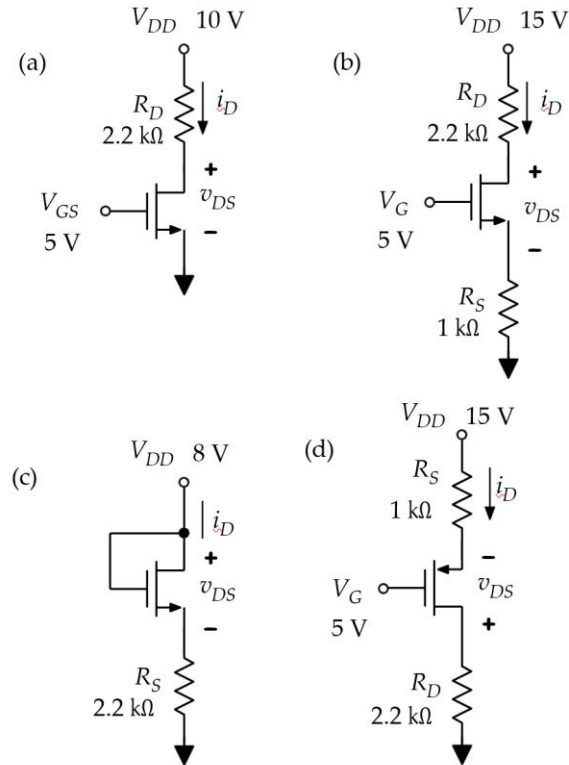
This transforms the equation into a linear relationship between $\sqrt{i_D}$ and v_{GS} :

- Slope $m = \sqrt{K}$
- X-intercept = V_T

The values for V_T and K are obtained from the square-root method and are reasonably close to those obtained from the parameter analyzer. However, the analyzer provided more accurate results with respect to physical devices and their limitations.

A DC MOS Circuit

After characterizing an n- and p-channel device, the below circuits were built, measured, and analyzed.



Circuit 1

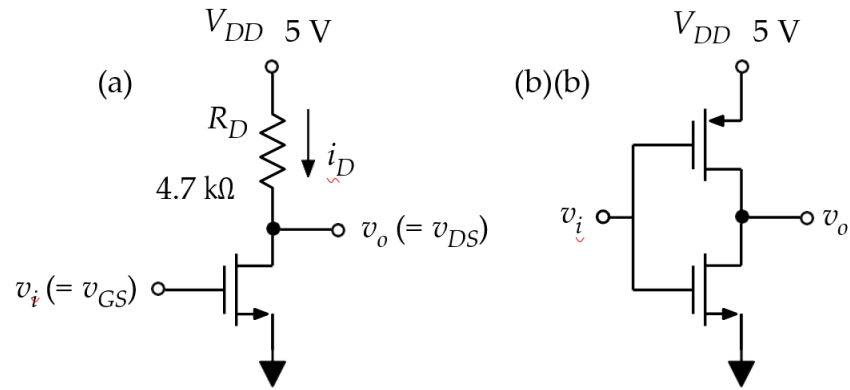
The table below shows the measured and calculated voltages and currents for each circuit.

Circuit	Calculated			Measured		
	i_D	v_{GS}	v_{DS}	i_D	v_{GS}	v_{DS}
A	3.89 mA	5 V	1.43 V	4 mA	5.01 V	1.53 V
B	1.76 mA	3.24 V	9.37 V	1.1 mA	3.88 V	11.41 V
C	2.08 mA	3.42 V	3.42 V	1.3 mA	5.16 V	5.16 V
D	4.35 mA	-6.1 V	-1.08 V	3.89 mA	-6.1 V	-2.45 V

Among the given circuits, circuit A operates in the triode region, circuit B and circuit C operate in the saturation region, and circuit D operates in the triode region too. From my calculations for circuit C, there is a clear discrepancy between the calculated and measured values for V_{GS} and V_{DS} . This was due to us choosing the wrong node when we were measuring the values.

B MOS Inverter

Two applications of a MOSFET as an inverter were then explored. The following two circuits were used:



Circuit 2

The transfer characteristic of each circuit was measured by sweeping the input voltage and measuring the output. These are shown below.

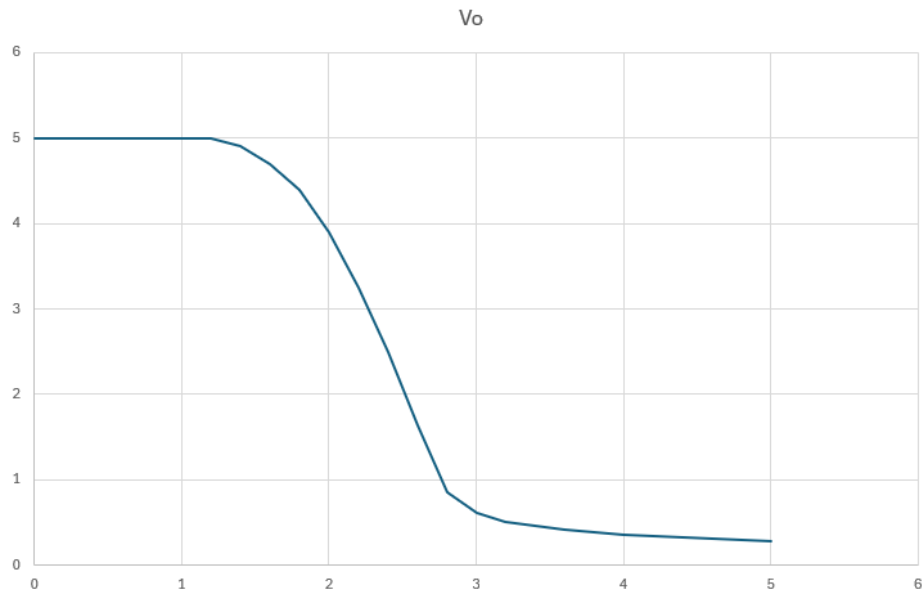


Figure 5: V_o vs V_{in} of Circuit A

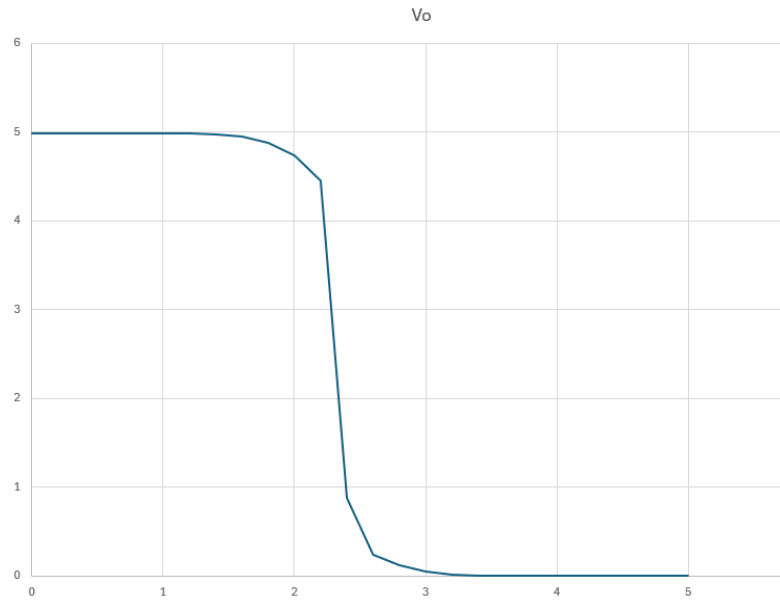


Figure 6: V_o vs V_{in} of Circuit B

Both inverter circuits showed the expected inverting behavior: when v_i was low, v_o was close to 5V, and as v_i increased, v_o decreased.

For the resistively loaded NMOS inverter (Circuit A), the transition was gradual, and the output did not reach 0V (around 0.3V). This is because the resistor and NMOS form a voltage divider when the transistor is on.

For the CMOS inverter (Circuit B), the transition was much sharper, and the output went from nearly 5V to nearly 0V. This happens because one transistor is always off while the other is on, pulling the output strongly to either V_{DD} or ground. Both circuits had similar maximum output voltages (around 5 V), but the CMOS inverter had a lower minimum output voltage and a sharper switch.

When v_i is high, the NMOS inverter uses more power because there is continuous current through the resistor and transistor. The CMOS inverter uses much less power since there is no direct path from V_{DD} to ground in steady state.

C MOSFET as a Switch

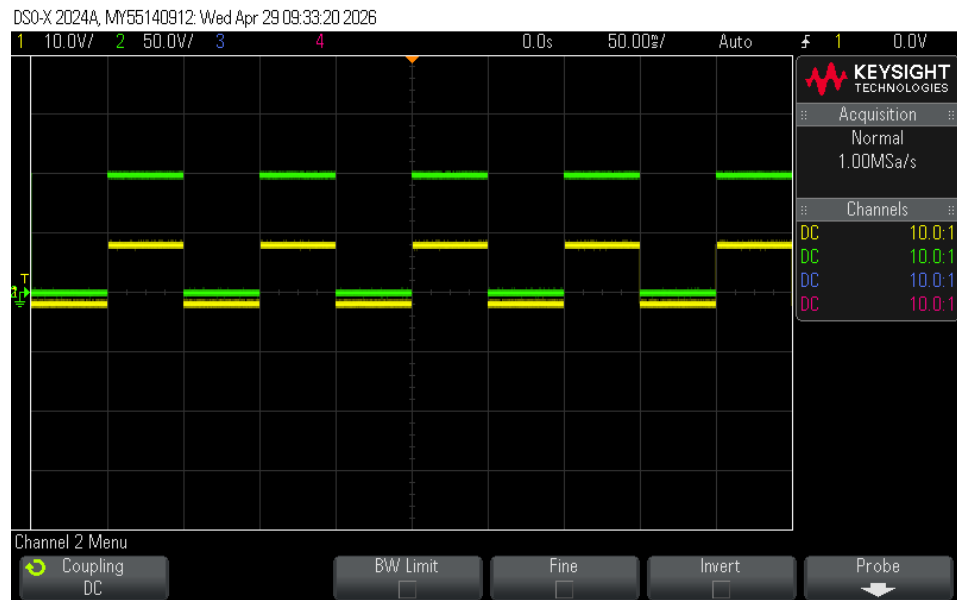


Figure 7: Oscilloscope trace of v_{DS}

D NMOS Amplifier

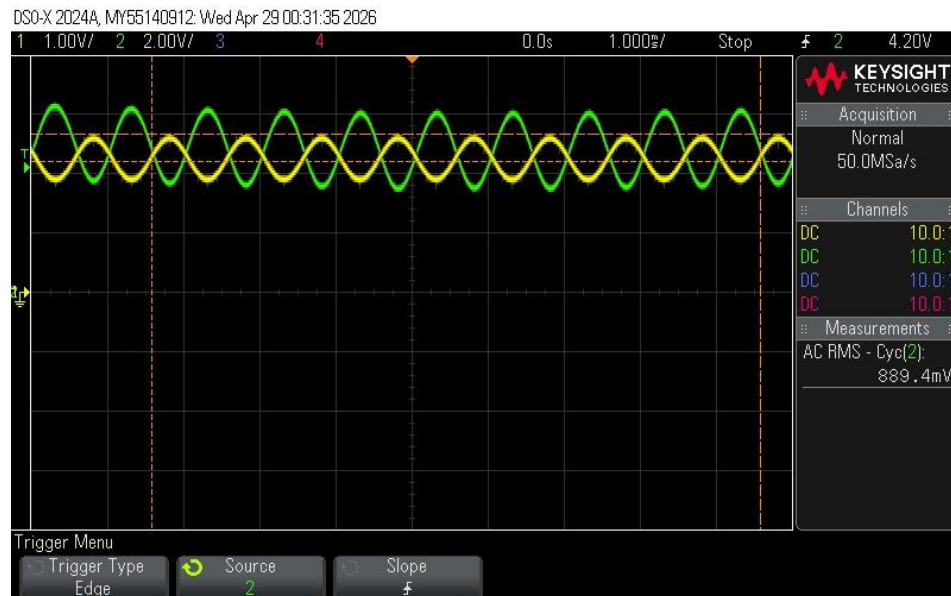


Figure 8: Oscilloscope trace of output and input voltages

In this circuit, the NMOS inverter is being used as an amplifier by biasing it in its linear (active) region. The DC offset is adjusted so that V_{DS} is around 5 V, which places the transistor around the middle of its operating curve. This allows small changes in the input voltage to produce larger changes in the output voltage.

When the AC input signal is applied, it slightly varies v_{GS} around this bias point. Because the transistor is operating in a region where its current changes significantly with v_{GS} , these small input variations cause larger changes in the drain current i_D . This, in turn, produces larger voltage swings across the drain resistor, resulting in an amplified output.

The output waveform is inverted relative to the input. This happens because an increase in v_{GS} increases i_D , which increases the voltage drop across the resistor and therefore reduces v_o . So, when the input goes up, the output goes down, and vice versa.

From the oscilloscope trace, the output signal clearly has a larger amplitude than the input, confirming amplification. The sinusoidal shape is preserved, which shows the circuit operating in a reasonably linear region around the bias point.

Conclusion

We didn't face any major issues during the lab. Almost all our calculated values closely agreed with our measured ones. Though, for circuit C in experiment A, our calculated values were a bit smaller than the measured values for V_{GS} and V_{DS} , but that was due to us getting the wrong node when measuring.